Precept 2: Warm Up Exercise

September 22, 2026

Problem

Consider the following vectors:

$$\vec{v} = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}, \qquad \vec{u}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \qquad \vec{u}_2 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

Let's answer the following questions:

1. What is the relationship between $\vec{u}_1$ and $\vec{u}_2$?
2. What is the dimension of the subspace spanned by $\vec{u}_1$ and $\vec{u}_2$?
3. How would you project vector $\vec{v}$ on to the subspace spanned by $\vec{u}_1$ and $\vec{u}_2$?

Remember the formula for vector projection is $\text{proj}_{\vec{u}}(\vec{v}) = \frac{\vec{v}\cdot\vec{u}}{\vec{u}\cdot\vec{u}}\vec{u}$

Solution

Question 1

$\vec{u}_1$ and $\vec{u}_2$ are orthogonal because

$$\vec{u}_1 \cdot \vec{u}_2 = 1 \cdot 1 + 2 \cdot 1 + 3 \cdot (-1) = 3 - 3 = 0$$

Question 2

Note that nonzero orthogonal vectors are automatically linearly independent. Hence, the subspace is a 2-dimensional subspace because there are two vectors and they are linearly independent.

Question 3

Because $\vec{u}_1$ and $\vec{u}_2$ are orthogonal, we can use the vector projection formula on both $\vec{v}$ on to $\vec{u}_1$ and $\vec{v}$ on to $\vec{u}_2$, and then add the resulting projections. Mathematically, this looks like

$$\frac{\vec{v} \cdot \vec{u}_1}{\vec{u}_1 \cdot \vec{u}_1}\vec{u}_1 + \frac{\vec{v} \cdot \vec{u}_2}{\vec{u}_2 \cdot \vec{u}_2}\vec{u}_2$$

which evaluates to

$$\frac{3}{7}\vec{u}_1 + \frac{4}{3}\vec{u}_2 = \frac{1}{21}\begin{bmatrix} 37 \\ 46 \\ -1 \end{bmatrix}$$

Note that this would not work if $\vec{u}_1$ and $\vec{u}_2$ were not orthogonal.